Microbiology 1

Science

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South East Technological University

Microbiology 1 (A07553)

Short Title:	Microbiology 1		
Faculty	Science and Computing		
Department	Science		
Cluster	Biology		
Author	AHEARNE		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module will describe basic microbiology, covering the biology of micro-organisms and the techniques used to study them.

Programmes

BIOL-0023	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	2 / 3 / M
BIOL-0023	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	2 / 3 / M
BIOL-0023	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	2 / 3 / M
BIOL-0023	BSc in Food Science with Business (WD_SFDSC_D)	2 / 3 / M
BIOL-0023	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	2 / 3 / M

Indicative Content

- Structure and reproduction of major groups of microbes: bacteria, fungi, viruses, algae and protozoa
- Identification of common bacteria
- Identification of common fungi
- Cultivation of microbes
- Microbial growth and its measurement
- Control of growth: disinfection, sterilisation
- Micro-organisms and diseases of plants and animals
- Control of disease: prevent transmission, vaccination, use of anti-microbial drugs

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Use standard laboratory techniques to isolate, purify, culture and identify bacteria as well as to quantify microbial numbers in cultures.
2. Describe the structure of major micro-organisms.
3. Recognise common bacteria and fungi as well as the microbiological media which can be used in their identification.
4. Identify appropriate sterilisation methods for media and materials.
5. Describe how the environment affects the growth of micro-organisms.
6. Estimate microbial numbers in cultures.
7. Outline the mechanisms by which disease is caused.
8. Record the results of laboratory work using worksheets.

Learning and Teaching Methods

- Lectures.

- Practical classes (incorporating lab based assessment / project work which is self-directed).

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	2, 3, 4, 5, 6, 7
Continuous Assessment	50%	
Practical	50%	1,8

Assessment Criteria

<40%: Little or no demonstration of an understanding of the fundamentals of microbiology. Lack of competence with associated laboratory skills & practices & an inability to carry out critical thinking.

40%-49%: Will show a limited understanding of the microbiology covered. May have some difficulty with applying critical thinking and problem-solving. Will display basic ability with associated laboratory skills & practices.

50%-59%: As well as the above, the student will show a good understanding of microbiology with demonstration of some critical thinking & problem-solving skills. Will display good competence with associated laboratory skills & practices.

60%-69%: In addition to the above, will demonstrate more detailed knowledge of microbiology, very good critical thinking skills & a high level of competence & efficiency in associated laboratory practicals.

70%-100%: All of the above to an excellent level.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- Madigan, M.T., J.M. Martinko, D.A. Stahl, D. Clark, K.S. Bender and Buckley, D.H. _Brock's Biology of Micro-organisms_. 13th Edition. Upper Saddle River, NJ: Prentice Hall / Pearson Education, 2011.

Supplementary Material

- "Moodle ." www.moodle.wit.ie
- Tortora, G.J., B.R. Funke and C.L. Case. _Microbiology - An Introduction_. 10th Edition. San Francisco: Pearson, 2010.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Biology Lab